
Exam 1

Grant Talbert

March 5, 2025

MA 542 Section A1

Boston University

Problem 1

- Clearly, the multiplicative identity of $\mathcal{P}(S)$ is S . For any $A \in \mathcal{P}(S)$, we have $A \subseteq S$, and thus $A \cap S = A$. The additive identity is also clearly the empty set. For any $A \in \mathcal{P}(S)$, we have

$$A + \emptyset = (A \cup \emptyset) \setminus (A \cap \emptyset) = A \setminus \emptyset = A.$$

- Since $A + (-A) = 0$, we have

$$(A \cup (-A)) \setminus (A \cap (-A)) = \emptyset.$$

Note that letting $A = -A$ satisfies this requirement:

$$A \cup A = A, \quad A \cap A = A, \quad \implies \quad (A \cup A) \setminus (A \cap A) = A \setminus A = \emptyset.$$

Therefore, $-A = A$.

- Since $1_{\mathcal{P}(S)} = S$ and $S = -S$, we have $S + S = 0$. Therefore, the additive order of the unital element is 2, and $\text{char } \mathcal{P}(S) = 2$.
- Note that the ring is clearly commutative for any S , as $A \cup B = B \cup A$ and likewise for the intersection, for any $A, B \subseteq S$. Moreover, $S \in \mathcal{P}(S)$ always exists, so the ring is always unital. Thus, the question of whether $\mathcal{P}(S)$ is an integral domain amounts to whether it has zero divisors.

If $S = \emptyset$, then $\mathcal{P}(S) = \{\emptyset\}$ is a singleton. It follows that $A, B \in \mathcal{P}(S)$ implies $A = B = S = 0$. It follows that there exist no zero divisors, since there are no nonzero elements as $\mathcal{P}(S)$ is the trivial ring. Therefore, $\mathcal{P}(S)$ is an integral domain.

If $S = \{*\}$ is a singleton, then $\mathcal{P}(S) = \{\emptyset, \{*\}\}$. The only nonzero element is $\{*\} = S$, and since S is the multiplicative identity, we have $S \cdot S = S \neq 0$, and thus S has no zero divisors. Therefore, $\mathcal{P}(S)$ is an integral domain.

If $|S| \geq 2$, then S contains at least two distinct elements $\mathbf{1}, \mathbf{2} \in S$. We then have $\{\mathbf{1}\}, \{\mathbf{2}\} \in \mathcal{P}(S)$ are distinct sets. Since $\mathbf{1} \neq \mathbf{2}$, we have

$$\{\mathbf{1}\} \cdot \{\mathbf{2}\} = \{\mathbf{1}\} \cap \{\mathbf{2}\} = \emptyset = 0.$$

Therefore, zero divisors exist, and $\mathcal{P}(S)$ is not an integral domain. Therefore, $\mathcal{P}(S)$ is an integral domain if and only if S is empty or is a singleton.

Problem 2

Let R be a finite commutative ring with unity, and let $\mathfrak{p} \subseteq R$ be a prime ideal. Since $\mathfrak{p}$ is prime, the quotient $R/\mathfrak{p}$ is an integral domain. Since $R/\mathfrak{p}$ is a quotient, there exists a natural projection $\pi : R \rightarrow R/\mathfrak{p}$ which is clearly surjective, as for any $r + \mathfrak{p} \in R/\mathfrak{p}$, there exists $r \in R$ such that $\pi(r) = r + \mathfrak{p}$. Since π is surjective, the cardinalities must obey $|R/\mathfrak{p}| \leq |R|$, and thus $R/\mathfrak{p}$ is finite. Since all finite integral domains are fields, we have that $R/\mathfrak{p}$ is a field. Since any quotient R/I is a field if and only if I is maximal in R , and $R/\mathfrak{p}$ is a field, we have that $\mathfrak{p}$ is maximal in R .

Problem 3

Since $a^i \in R$ for all $i \in \{0, \dots, n\}$, given that $a^0 = 1$, we have that

$$\sum_{i=0}^n a^i \in R.$$

Consider the product of this element with the element $(1 - a)$:

$$\begin{aligned} (1 - a) \left(\sum_{i=0}^n a^i \right) &= \sum_{i=0}^n a^i - a \sum_{i=0}^n a^i \\ &= \sum_{i=0}^n a^i - \sum_{i=0}^n a^{i+1} \\ &= \sum_{i=0}^n a^i - \sum_{i=1}^{n+1} a^i \\ &= \left(1 + \sum_{i=1}^n a^i \right) - \left(\sum_{i=1}^n a^i \right) - a^{n+1} \\ &= 1 + \left(\sum_{i=1}^n a^i - \sum_{i=1}^n a^i \right) - a^{n+1} \\ &= 1 - a^{n+1} \\ &= 1 - a^n a \\ &= 1 - 0a \\ &= 1. \end{aligned}$$

$$\begin{aligned} \left(\sum_{i=0}^n a^i \right) (1 - a) &= \sum_{i=0}^n a^i - \left(\sum_{i=0}^n a^i \right) a \\ &= \sum_{i=0}^n a^i - \sum_{i=0}^n a^{i+1} \\ &\quad \text{(same math as above)} \\ &= 1. \end{aligned}$$

Therefore, the sum $\sum_{i=0}^n a^i$ is an inverse of $(1 - a)$.

Problem 4

Factorization into irreducibles in a UFD is unique up to associates and number of irreducibles. Note that since $2 \in \mathbb{Z}_7[x]$ and $2(2x + 5) \equiv 4x + 3 \pmod{7}$, the polynomials $2x + 5$ and $4x + 3$ are associates. Additionally, $3, 4, 5 \in \mathbb{Z}_7[x]$, and we have

$$3(2x + 3) = 6x + 2, \quad 4(3x + 2) = 5x + 1, \quad 5(3x + 2) = x + 3.$$

Therefore, all elements on the left are associates of elements on the right. This means they are equivalent factorizations up to associates, and thus not in violation of the fact that $\mathbb{Z}_7[x]$ is a UFD.

Problem 5

- Since Q is prime, $ab \in Q$ implies that $a \in Q$ or $b \in Q$. Therefore, if $a, b \in R \setminus Q = S$, then $a, b \notin Q$, and thus $ab \notin Q$ by contrapositive. Since $ab \in R$, we have $ab \in S$.
- Since Q is a prime ideal, $Q \subsetneq R$. It follows that there exists $r \in S$, and since $0 \in Q$ since Q is an ideal and thus a subring, we have that $r \neq 0$. Since R is a domain, there exists $1 \in R$, and thus

$$\frac{1}{r} \in R_Q.$$

Therefore, R_Q is nonempty. Note that it is also closed under subtraction: for all $x/y, a/b \in R_Q$,

$$\frac{x}{y} - \frac{a}{b} = \frac{xb}{yb} - \frac{ya}{yb} = \frac{xb - ya}{yb}.$$

$xb - ya \in R$ obviously, since R is a ring. Since $y, b \in S$, by part 1 we have that $yb \in S$. Therefore, the resulting fraction is an element of R_Q . This makes R_Q into a group. Finally, R_Q is closed under multiplication: for any $x/y, a/b \in I$,

$$\frac{x}{y} \frac{a}{b} = \frac{xa}{yb}.$$

By the same logic, $xa \in R$ and $yb \in S$. It follows that R_Q is a subring of $\text{Frac}(R)$ by the subring test or whatever that theorem was called.

- Since Q is an ideal, it must be a subring, and thus $0 \in Q$. It follows that for any $x \in S$, $0/x = 0 \in I$, and thus I is nonempty. Now note that I is closed under subtraction: for any $x/y, a/b \in I$,

$$\frac{x}{y} - \frac{a}{b} = \frac{xb - ya}{yb}.$$

Since $y, b \in S$, by part 1 we have $yb \in S$. Since $x, a \in Q$, and since Q is an ideal, we have $xb, ya \in Q$, and thus $xb - ya \in Q$ since it is a subring and thus closed under addition. Therefore, I is closed under subtraction. Finally, let $x/y \in I$ and $a/b \in R_Q$. We have

$$\frac{x}{y} \frac{a}{b} = \frac{xa}{yb}.$$

Since $y, b \in S$, we have $yb \in S$. Since Q is an ideal and $x \in Q$, we have $xa \in Q$, and thus $xa/yb \in I$. Therefore, I attains the absorption property of ideals, and is automatically thus closed under multiplication. Therefore, I is an ideal.

- Let $I \subsetneq J$. Then there exists $x/y \in J \setminus I$. Since $x/y \in R_Q \setminus I$, we have that $y \in S$ by definition of R_Q , and $x \in R \setminus Q = S$. Therefore, $x, y \in S$. Since $x \notin Q$, and $0 \in Q$, we have $x \neq 0$. It follows that there exists $y/x \in R_Q$ since $x, y \in S \subseteq R$, and

$$\frac{x}{y} \frac{y}{x} = 1_{R_Q}.$$

Since $x/y \in J$, we have $1_{R_Q} \in J$. By logic that has been discussed at length previously, we have that $J = R_Q$. Therefore, I is maximal.

Problem 6

- Clearly, the additive and multiplicative identities are $0\mathbf{e} + 0\mathbf{g} + 0\mathbf{g}^2$ and $1\mathbf{e} + 0\mathbf{g} + 0\mathbf{g}^2$, respectively. To show this, let $a\mathbf{e} + b\mathbf{g} + c\mathbf{g}^2 \in \mathbb{Z}[G]$. We have

$$(a + 0)\mathbf{e} + (b + 0)\mathbf{g} + (c + 0)\mathbf{g}^2 = a\mathbf{e} + b\mathbf{g} + c\mathbf{g}^2.$$

We also have

- Note that in any ring R , if a is a unit and $\phi : R \rightarrow S$ is a ring homomorphism that preserves unity, then the images $\phi(a), \phi(a^{-1})$ must also be units:

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(1_R) = \phi(1_S) \implies \phi(a^{-1}) = \phi(a)^{-1}.$$

Note also that ε preserves unity; that is,

$$\varepsilon(1\mathbf{e}) = 1.$$

Therefore, units in $\mathbb{Z}[G]$ must map to a unit in $\mathbb{Z}$. It is a well-known fact that the only units in $\mathbb{Z}$ are ± 1 , so we must have $\varepsilon(a) = \pm 1$ if a is to be a unit.

- Let $\mathbf{a} := a_0\mathbf{e} + a_1\mathbf{g} + a_2\mathbf{g}^2$ and $\mathbf{b} := b_0\mathbf{e} + b_1\mathbf{g} + b_2\mathbf{g}^2 \in \mathbb{Z}[G]$. Then

$$\varepsilon(\mathbf{a}) + \varepsilon(\mathbf{b}) = (a_0 + a_1 + a_2) + (b_0 + b_1 + b_2) = (a_0 + b_0) + (a_1 + b_1) + (a_2 + b_2) = \varepsilon(\mathbf{a} + \mathbf{b}).$$

$$\begin{aligned} \varepsilon(\mathbf{a})\varepsilon(\mathbf{b}) &= (a_0 + a_1 + a_2)(b_0 + b_1 + b_2) \\ &= (a_0b_0 + a_0b_1 + a_0b_2) + (a_1b_0 + a_1b_1 + a_2b_1) + (a_2b_0 + a_2b_1 + a_2b_2) \\ &= (a_0b_0 + a_1b_2 + a_2b_1) + (a_0b_1 + a_1b_0 + a_2b_2) + (a_0b_2 + a_1b_1 + a_2b_0) \\ &= \varepsilon(\mathbf{ab}). \end{aligned}$$

Therefore, ε is a ring homomorphism.

- Note that for all $n \in \mathbb{Z}$, $n\mathbf{e} \in \mathbb{Z}[G]$, and $\varepsilon(n\mathbf{e}) = n$. Therefore, ε is surjective. By the first isomorphism theorem,

$$\frac{\mathbb{Z}[G]}{\ker \varepsilon} \cong \text{im } \varepsilon = \mathbb{Z}.$$

- The polynomial in question is $f(x) = x^3 - 1$. This is fairly obvious, since we want $x^3 = 1$ in $\mathbb{Z}[G]$, so modding out by the relation gives the ring in question. The homomorphism $x \mapsto g$ also exists since $\mathbb{Z}[x]$ is free as a $\mathbb{Z}$ -algebra.

•

$$(\mathbf{e} + \mathbf{g} + \mathbf{g}^2)(\mathbf{e} - \mathbf{g}) = (1 - 1)\mathbf{e} + (1 - 1)\mathbf{g} + (1 - 1)\mathbf{g}^2 = 0.$$

Since both of these are elements of $\mathbb{Z}[G]$, they are zero divisors, and thus $\mathbb{Z}[G]$ is not an integral domain, as it has zero divisors.

- Note that there exists an inclusion map $D_3 \hookrightarrow \mathbb{Z}[D_3]$, given by mapping $x \mapsto 1x$. Note that $rs \neq sr$ in D_3 for r the rotation and s the reflection, as can be verified by explicit calculation.

I will prove a more general case, because I feel like it. Suppose that there exists any ideal $I \subseteq \mathbb{Z}[x]$ such that

$$\frac{\mathbb{Z}[x]}{I} \cong \mathbb{Z}[D_3].$$

Suppose this isomorphism is given by a function ϕ . Since $\mathbb{Z}[x]$ is commutative and since there exists a canonical projection $\mathbb{Z} \rightarrow \mathbb{Z}/I$ which is surjective, by properties of homomorphisms $\mathbb{Z}[x]/I$ is commutative. Since ϕ is an isomorphism, there exist unique $a, b \in \mathbb{Z}[x]/I$ such that $\phi(a) = r$, $\phi(b) = s$. We have

$$rs = \phi(a)\phi(b) = \phi(ab) = \phi(ba) = \phi(b)\phi(a) = sr.$$

This contradicts the fact that $rs \neq sr$. Therefore, there exists no quotient of $\mathbb{Z}[x]$ satisfying this isomorphism. This includes quotients by principal ideals.